

Element D

Brainstorming Ideas and Mechanism Selection

We decided to break our design down into smaller mechanisms and systems before combining everything together into the final X-Ray holder. This helped us compare simpler ideas first before deciding on the larger overall design!!

1. Corner Piece Mechanism

Decision Matrix

Brainstormed Ideas

- 3D Printed PLA Corner Pieces
- Wooden Corner Blocks
- Aluminum Corner Brackets
- Fully Glued Polycarbonate Edges

Design	Advantages	Disadvantages
3D Printed PLA Pieces	Precise measurements, lightweight, easy to redesign	Less durable than metal
Wooden Blocks	Cheap and strong	Harder to shape precisely
Aluminum Brackets	Very strong	Heavy and expensive
Glued Edges	Simple and lightweight	Weak and hard to repair

The 3D printed corner pieces scored the highest because they allowed very exact measurements and could easily be reprinted if dimensions changed. Since our measurements changed several times during the project, this made 3D printing the best choice.

Category	Importance %	PLA Corner Pieces	Wooden Blocks	Aluminum Brackets	Glued Edges
Precision	25%	10	6	8	5
Ease of Manufacturing	20%	9	7	5	8
Weight	15%	9	6	5	10
Durability	20%	7	8	10	4
Cost	10%	8	9	6	10
Adjustability	10%	9	5	6	2
Total	100%	8.55	6.85	7.15	5.90

2. Latch Mechanism Decision Matrix

Brainstormed Ideas

- Chest Latches
- Magnetic Latch
- Hook-and-Eye Latch
- Sliding Lock
- Velcro Closure

Design	Advantages	Disadvantages
Chest Latches	Strong and secure	Slightly heavier
Magnetic Latch	Easy to open	Not strong enough
Hook-and-Eye	Cheap and simple	Slower to use
Sliding Lock	Durable	Harder to align
Velcro Closure	Fast and lightweight	Weak over time

The chest latches worked best because they can securely hold the X-Ray holder shut while still being easy to open and close. Other options like Velcro or magnets were easier to use but were not secure enough for the expensive X-Ray plate.

Category	Importance %	Chest Latches	Magnetic Latch	Hook-and-Eye	Sliding Lock	Velcro Closure
Security	30%	10	6	7	8	4
Ease of Use	20%	9	10	6	7	10
Durability	20%	9	5	8	8	4
Safety	15%	8	9	7	8	8
Cost	15%	8	7	10	7	10
Total	100%	8.95	7.20	7.45	7.65	6.50

3. Clamp-to-Plate Connection Decision Matrix

Brainstormed Ideas

- C-Clamp Connection
- Bolted Polycarbonate Frame
- Magnetic Holder
- Velcro Strap Holder
- Sliding Rail Holder

Design	Advantages	Disadvantages
C-Clamp	Easy to attach	Can damage plate edges
Bolted Polycarbonate Frame	Strong and protective	More difficult to build
Magnetic Holder	Quick access	Weak support
Velcro Holder	Lightweight	Less durable
Sliding Rail Holder	Adjustable	More complicated

The bolted polycarbonate frame scored the highest because it best protected the expensive X-Ray plate while still remaining strong and secure. It also reduced the chances of the plate falling out or moving during X-Rays.

Category	Importance %	C-Clamp	Bolted Polycarbonate Frame	Magnetic Holder	Velcro Strap Holder	Sliding Rail Holder
Security of X-Ray Plate	30%	7	10	5	6	8
Safety	20%	6	9	7	8	8
Ease of Access	15%	7	8	10	9	6
Durability	15%	8	9	5	5	8
Ease of Manufacturing	10%	8	7	6	10	5
Cost	10%	9	7	5	10	6
Total	100%	7.35	8.70	6.20	7.45	7.10

4. Adjustable Arm / Wall Monitor Decision Matrix

Brainstormed Ideas

- Wall Monitor Mount Arm
- Camera Tripod Arm
- Hydraulic Arm
- PVC Pipe Arm
- Flexible Gooseneck Arm

Design	Advantages	Disadvantages
Monitor Arm	Adjustable and strong	Slight wobbling
Camera Tripod Arm	Lightweight	Not strong enough
Hydraulic Arm	Very smooth movement	Expensive
PVC Pipe Arm	Cheap	Hard to adjust
Gooseneck Arm	Flexible	Sags under weight

Category	Importance %	Monitor Arm	Camera Tripod Arm	Hydraulic Arm	PVC Arm	Gooseneck Arm
Adjustability	25%	10	8	10	4	9
Strength	25%	9	6	10	5	4
Ease of Attachment	15%	9	7	5	6	9
Durability	15%	8	6	10	5	5
Cost	10%	8	7	3	10	7
Weight	10%	8	8	3	9	9
Total	100%	8.85	6.95	7.55	6.10	6.65

The monitor wall mount arm worked best because it provided strong support while still allowing movement in multiple directions. The other stuff we researched like Hydraulic arms etc. also could work well but were too expensive/complicated for a high school engineering project.

5. Pole Clamp Mechanism Decision Matrix

Brainstormed Ideas

- Ratchet Strap Clamp
- Bolt Tightened Wood Clamp
- Metal U-Bolt Clamp
- Velcro Strap Clamp
- Rope Tie System

Design	Advantages	Disadvantages
Clamp + Monitor Arm	Stable and adjustable	More parts to assemble
Sliding Rail	Very stable	Bulky
Gooseneck Arm	Flexible	Weak under heavy load
Freestanding Tripod	Independent structure	Zoo did not want freestanding

Category	Importance %	Ratchet Strap Clamp	Bolt Tightened Clamp	Metal U-Bolt Clamp	Velcro Strap Clamp	Rope Tie System
Stability	30%	9	8	10	5	4
Ease of Setup	20%	9	5	4	10	6
Safety Around Animals	15%	9	7	6	9	7
Adjustability	15%	10	6	5	9	8
Durability	10%	8	9	10	5	4
Cost	10%	8	8	6	9	10
Total	100%	8.95	6.95	6.90	7.65	5.85

The ratchet strap clamp scored the highest because it was strong, adjustable, and easy to attach to different enclosure poles. It also allowed quick setup and safer movement if giraffes accidentally pushed the holder.

6. Pole Clamp + Adjustable Arm Combination Decision Matrix

Brainstormed Ideas

- Ratchet Strap Clamp + Monitor Arm
- Sliding Rail System
- Flexible Gooseneck Arm
- Freestanding Tripod

Design	Advantages	Disadvantages
Clamp + Monitor Arm	Stable and adjustable	More parts to assemble
Sliding Rail	Very stable	Bulky
Gooseneck Arm	Flexible	Weak under heavy load
Freestanding Tripod	Independent structure	Zoo did not want freestanding

Category (Rated 1–10)	Importance %	Clamp + Monitor Arm	Sliding Rail System	Gooseneck Arm	Freestanding Tripod
Stability	25%	9	10	5	6
Safety	20%	9	8	6	5
Adjustability	15%	10	7	9	8
Durability	15%	8	9	5	7
Ease of Setup	10%	8	6	10	7
Portability	10%	8	5	9	5
Cost	5%	8	6	7	5
Total	100%	8.75	7.95	6.55	6.10

The ratchet strap clamp combined with the monitor arm gave the best balance between stability, movement, and portability. It also matched the zoo's request for a design like this!

7. Overall Medical X-Ray Holder Design Decision Matrix

Brainstormed Overall Designs

- Zoo Example Medical X-Ray Holder
- Our Pole Clamp and Polycarbonate Design
- Sliding Rail Design
- Freestanding Tripod Design
- Flexible Arm Design

Design	Advantages	Disadvantages
Zoo Example Holder	Professional and stable	Expensive and bulky
Our Final Design	Portable, adjustable, lower cost	Less polished
Sliding Rail Design	Stable	Heavy and hard to transport
Freestanding Tripod	Easy placement	Unsafe around giraffes
Flexible Arm Design	Lightweight	Weak support

The medical X-Ray holder example given by the zoo helped us understand the overall goal, but it was not practical for the Nashville Zoo barns because it was too expensive, bulky, and designed more for hospitals. Our own final design scored the highest because it was specifically built for giraffe and okapi enclosures while staying portable, adjustable, and affordable.

Category	Importance %	Zoo Example Holder	Our Final Design	Sliding Rail	Freestanding Tripod	Flexible Arm
Compatibility with Zoo Request	25%	4	10	8	2	7
Portability	15%	5	8	5	4	9
Stability	20%	10	9	10	6	5
Safety Around Animals	15%	6	9	7	5	6
Ease of Setup	10%	6	8	5	7	10
Cost	10%	2	8	5	6	7
Ease of Manufacturing	5%	3	8	4	7	8
Total	100%	5.80	8.80	6.85	4.85	6.90

Final Design Selection

The ratchet strap pole clamp with adjustable monitor arm received the highest overall score in our decision matrix and was selected as the final design!! This design had balanced stability, portability, adjustability, safety, durability, and a realistic building process within our classroom.

The design also aligned closely with stakeholder feedback from zoo veterinarians/zookeepers. The monitor arm allowed enough movement and positioning for X-Ray procedures while the clamp system securely attached to existing enclosure poles. Additionally, the design allowed slight movement instead of remaining completely rigid, reducing the chance of damage if a giraffe accidentally pushed against the assembly.

Design Blueprint (D4)

Final Design Overview

The final prototype consists of four major systems:

1. Pole Clamp System
2. Adjustable Monitor Arm
3. Polycarbonate X-Ray Holder
4. Corner Pieces and Latch Mechanism

The pole clamp attaches to square enclosure poles using ratchet straps. A monitor arm is attached to the center of the clamp and supports the polycarbonate X-Ray holder. The holder protects and secures the X-Ray plate while allowing movement and positioning.

Pole Clamp System Analysis

The pole clamp system uses two wood pieces connected and tightened around enclosure poles using ratchet straps. Hot glue was added to the back of the clamp to increase friction and reduce slipping.

Materials

- Classroom wood
- Ratchet straps
- Wood glue
- Hot glue
- Bolts and washers

Manufacturing Methods

1. Measure and cut wood pieces.
2. Sand rough edges.
3. Glue and assemble wood pieces.
4. Attach ratchet straps.
5. Add hot glue patterns for friction.

Adjustable Monitor Arm Analysis

A monitor wall mount arm connects the pole clamp to the X-Ray holder and allows vertical and rotational movement.

Materials

- Monitor wall mount arm
- Screws
- Bolts and washers

Manufacturing Methods

1. Measure center placement on clamp.
2. Drill mounting holes.
3. Attach monitor arm using screws and bolts.

Analysis

The monitor arm can provide strong movement and adjustability while supporting the weight of the X-Ray holder. It can allow the holder to rotate and move vertically, which can be important for positioning the X-Ray plate.

Polycarbonate X-Ray Holder Analysis

The polycarbonate holder protects and secures the X-Ray plate while attached to the monitor arm.

Materials

- Polycarbonate sheets
- Bolts and nuts
- Washers
- Latch mechanisms
- 3D printed corner pieces

Manufacturing Methods

1. Design holder dimensions in CAD.
2. Cut polycarbonate sheets using CNC machine.
3. Drill holes for bolts and assembly.
4. Assemble sheets with corner pieces and latches.

Analysis

The polycarbonate holder can successfully protect the space for the X-Ray plate and allowed secure positioning. Multiple prototype versions can be necessary!

Corner Pieces and Latch Mechanism Analysis

The corner pieces and latch mechanism hold the polycarbonate sheets together and allow access to the inside of the holder.

Materials

- PLA filament
- Latches
- Bolts and nuts

Manufacturing/Our building Methods

1. Design parts using CAD software.
2. 3D print all components.
3. Assemble onto polycarbonate sheets.

Analysis

The 3D printed corner pieces and latch mechanism can help the holder to stay secured and also remain easy to open and close. Several prints can be needed to adjust dimensions after testing. Reprinting different size corner pieces can help solve spacing problems later on.

Supply List and Costs:

Material name	Description	Where material was purchased	Cost	# units
Bambu PLA	3D printing material (Bambu printer)	Bambu website	\$32.00	1
Latches	Matte Black Chest Door Latch	Home depot	3.77	2
Bolts/nuts	3 inch bolts/nuts	Home depot	\$20.00	20
Wall monitor	MOUNTUP Monitor Wall Mount	MOUNTUP/amazon	\$40.00	1
Ratchet straps	Retractable Ratchet Tie-Downs - S-Hook	POWERTYE MFG/amazon	\$12.00	2
Total			\$107.77	

Plan of Action (D3)

